DEPI GRADUATION PROJECT | FULL STACK .NET TRACK

4.5 System Deployment & Integration

MediCare — Clinic Management & Appointment System

Team 1
Submission: 6 Nov 2026

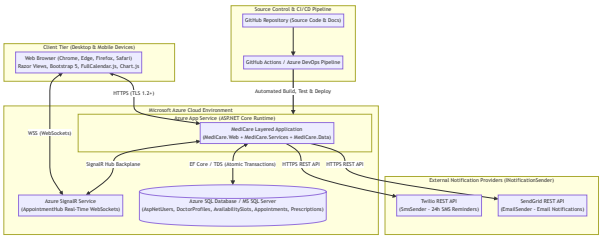
5.1 Technology Stack

The MediCare platform integrates frontend, backend, database, and cloud technologies structured around a **Layered MVC Monolith / N-Tier Architecture** (`MediCare.Web`, `MediCare.Services`, `MediCare.Data`):

Layer / Category	Selected Technologies	Architectural Role & Implementation Purpose
Frontend Technologies	`Razor Views`, `Bootstrap 5`, `JavaScript`, `FullCalendar.js`, `Chart.js`	Renders responsive, mobile-first interfaces adhering to the brand palette (`#56060D` burgundy, `#A81420` crimson, `#FFF9F1` ivory) and WCAG 2.2 AA accessibility guidelines; displays interactive booking calendars and administrative analytics charts.
Backend Technologies	`C#`, `ASP.NET Core MVC`, `ASP.NET Core Identity`, `SignalR`	Implements thin MVC controllers (`MediCare.Web`), role-based authorization policies (`Patient`, `Doctor`, `Admin`), real-time WebSocket hubs (`AppointmentHub`), and decoupled business services (`MediCare.Services`) using Dependency Injection and the Factory Pattern (`NotificationFactory`).
Database Technologies	`Microsoft SQL Server / Azure SQL Database`, `EF Core (Code-First)`	Manages normalized (3NF) relational tables (`AspNetUsers`, `DoctorProfiles`, `AvailabilitySlots`, `Appointments`, `Prescriptions`) and extended clinical entities via `IUnitOfWork`, `IGenericRepository<T>`, and specific repositories (`IDoctorRepository`, `IPatientRepository`, `IAppointmentRepository`, `IMedicalRecordRepository`).
Cloud & External Integrations	`Azure App Service`, `Azure SignalR Service`, `Twilio (SMS)`, `SendGrid (Email)`, `GitHub Actions`	Hosts the web application and real-time hubs in the cloud, dispatches SMS/Email confirmations and 24-hour reminders via `INotificationSender` (with In-App/Mock fallback), and automates CI/CD deployment pipelines.

5.2 Deployment Diagram

The Deployment Diagram illustrates how MediCare software components are distributed across client devices, Microsoft Azure cloud infrastructure, and external notification providers:



5.3 Component Diagram

The Component Diagram depicts the high-level software components across the three N-Tier layers (`MediCare.Web`, `MediCare.Services`, `MediCare.Data`) and their interface dependencies:

